

The finite-dimensional Khintchine constants on B_q^n for $p = 4$ and
 $p = 6$

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Source Problem

The source is Alexandros Eskenazis, Piotr Nayar, and Tomasz Tkocz, *Sharp comparison of moments and the log-concave moment problem*, arXiv:1801.07597. Their Question 5 asks:

Fix $q \in (2, \infty)$, $n \geq 1$, and let $X = (X_1, \dots, X_n)$ be a random vector uniformly distributed on B_q^n . For $p \geq 2$, what are the optimal values of $B_{p,q,n}$ in the finite-dimensional Khintchine inequality? More ambitiously, which unit vectors maximise the moments $\|\sum_{i=1}^n a_i X_i\|_p$?

The same question also asks for the dual missing lower constants when $p \in [1, 2]$, and the paper separately notes that $q > 2$, $p \in (-1, 1)$ remains open.

Corollary 4. Fix $q \in [2, \infty]$. If $n \geq 1$ and $X = (X_1, \dots, X_n)$ is a random vector uniformly distributed on B_q^n , then for every real numbers $a_1, \dots, a_n$ and $p \geq 1$ we have

$$(8) \quad A_p \left\| \sum_{i=1}^n a_i X_i \right\|_2 \leq \left\| \sum_{i=1}^n a_i X_i \right\|_p \leq B_p \left\| \sum_{i=1}^n a_i X_i \right\|_2,$$

where

$$(9) \quad A_p = \begin{cases} \gamma_p, & p \in [1, 2) \\ \frac{3^{1/2}}{(p+1)^{1/p}}, & p \in [2, \infty) \end{cases} \quad \text{and} \quad B_p = \begin{cases} \frac{3^{1/2}}{(p+1)^{1/p}}, & p \in [1, 2) \\ \gamma_p, & p \in [2, \infty) \end{cases}$$

The above constants are sharp.

It will be evident from the proof of Corollary 4 that the dimension-dependent constants (7) improve upon the asymptotically sharp constants given in (9).

Question 5. Fix $q \in (2, \infty)$, $n \geq 1$ and let $X = (X_1, \dots, X_n)$ be a random vector uniformly distributed on B_q^n . For $p \geq 2$ (respectively $p \in [1, 2)$), what are the optimal values of $B_{p,q,n}$ (resp. $A_{p,q,n}$) in (6)? More ambitiously, which unit vectors $(a_1, \dots, a_n)$ maximise (resp. minimise) the moments $\left\| \sum_{i=1}^n a_i X_i \right\|_p$?

Our arguments rely on the convexity of certain functions and work in fact for the whole range $q > 0$. However, when $p < 1$ those functions are no longer convex. The technique developed in [ENT16] for the range $q \in (0, 2]$ has the advantage of covering all $p > -1$. It remains an open problem to understand the optimal constants for $q > 2$ and $p \in (-1, 1)$.

1.2. An extremal property of symmetric uniform distributions. Before proceeding to the second main part of the present article, we mention an extremal property of symmetric uniform random variables which was motivated by a similar property of independent symmetric random signs $\varepsilon_1, \varepsilon_2, \dots$. In [FHJSZ97] and independently in [Pin94], the authors showed that an Orlicz function $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ of class C^2 satisfies the inequality

$$(10) \quad \mathbb{E} \Phi \left(\sum_{i=1}^n X_i \right) \geq \mathbb{E} \Phi \left(\sum_{i=1}^n \sigma_i \varepsilon_i \right),$$

for every symmetric independent random variables $X_1, X_2, \dots$ and real numbers $\sigma_1, \sigma_2, \dots$ such that $\sigma_i^2 = \mathbb{E} X_i^2$ if and only if Φ'' is convex on $\mathbb{R}$. This result, when applied to $\Phi(x) = |x|^p$ and X_i being standard Gaussian random variables allows one to derive the optimal constants in the classical Khintchine inequality for $p \geq 3$. For $p \in (0, 3)$ all available proofs (see [Haa81], [NP00], [Mor17]) are subtle and more technical. We obtain the following analogue of the above theorem for symmetric unimodal random variables, i.e. continuous random variables whose densities are even and nonincreasing on $[0, \infty)$.

Theorem 6. Let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[-\sqrt{3}, \sqrt{3}]$, thus having $\mathbb{E} U_i^2 = 1$. An even function $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ of class C^3 satisfies

$$(11) \quad \mathbb{E} \Phi \left(\sum_{i=1}^n X_i \right) \geq \mathbb{E} \Phi \left(\sum_{i=1}^n \sigma_i U_i \right),$$

for every symmetric unimodal independent random variables $X_1, X_2, \dots$ and real numbers $\sigma_1, \sigma_2, \dots$, such that $\sigma_i^2 = \mathbb{E} X_i^2$, if and only if $\Phi''(x) \geq 0$ for every $x \geq 0$. Moreover, the reverse inequality holds if and only if $\Phi''(x) \leq 0$ for every $x \geq 0$.

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Result

This packet solves the $p = 4$ and $p = 6$ cases of the missing finite-dimensional upper-constant problem for every $q \in (2, \infty)$ and every $n \geq 1$. The optimising directions are exactly the equal-modulus directions, $|a_1| = \dots = |a_n| = n^{-1/2}$.

Proof Intuition

The source paper reduces the uniform measure on B_q^n to sums of independent coordinates Y_i with density proportional to $e^{-|x|^q}$. For $p = 4$, the moment is

$$\mathbb{E} \left(\sum a_i Y_i \right)^4 = 3m_2^2 + (m_4 - 3m_2^2) \sum a_i^4,$$

where $m_r = \mathbb{E}Y^r$. Since the $q > 2$ coordinate law has fourth moment strictly below the Gaussian fourth moment after variance normalisation, the coefficient of $\sum a_i^4$ is negative. The maximum therefore occurs when $\sum a_i^4$ is as small as possible, i.e. at equal moduli.

For $p = 6$, the moment is still only a cubic symmetric polynomial in $x_i = a_i^2$. The same fourth-moment deficit gives one negative coefficient. A second inequality, $m_6 < 5m_4m_2$, obtained by applying the source paper's one-coordinate convex comparison with fourth-moment normalisation, makes the Schur-concavity derivative bracket negative on the whole simplex. Thus the sixth moment is strictly Schur-concave in a_i^2 , and the equal-modulus vector is again the unique maximiser up to signs.

Lemma 1 (Moment deficits). *Let $q > 2$, and let Y have density proportional to $e^{-|x|^q}$. Put $m_{2r} = \mathbb{E}Y^{2r}$. Then*

$$m_4 < 3m_2^2, \quad m_6 < 5m_4m_2.$$

Proof. Lemma 11 in the source paper says that if $0 < q_0 < q_1$, $r > 0$, and h is convex on $[0, \infty)$, then

$$\mathbb{E}h \left(\left| Y^{(q_0)} / \|Y^{(q_0)}\|_r \right|^r \right) \geq \mathbb{E}h \left(\left| Y^{(q_1)} / \|Y^{(q_1)}\|_r \right|^r \right),$$

where $Y^{(s)}$ has density proportional to $e^{-|x|^s}$. Take $q_0 = 2$, $q_1 = q$, $r = 2$, and $h(t) = t^2$. Since the $q_0 = 2$ law is Gaussian, this gives $m_4/m_2^2 < 3$ for $q > 2$. The inequality is strict here, and below, because the normalised $q = 2$ and $q > 2$ laws are distinct and the test functions are strictly convex on the support.

Next take $r = 4$ and $h(t) = t^{3/2}$. This gives

$$\frac{m_6}{m_4^{3/2}} < \frac{\mathbb{E}G^6}{(\mathbb{E}G^4)^{3/2}} = \frac{15}{3^{3/2}} = \frac{5}{\sqrt{3}},$$

where G is centred Gaussian. Combining this with $m_4 < 3m_2^2$, we get

$$m_6 < \frac{5}{\sqrt{3}}m_4^{3/2} < 5m_4m_2.$$

□

Theorem 1 (The missing constants for $p = 4$ and $p = 6$). *Let $q \in (2, \infty)$, $n \geq 1$, and let X be uniformly distributed on B_q^n . Let Y have density proportional to $e^{-|x|^q}$, put $m_{2r} = \mathbb{E}Y^{2r}$, and set $\rho_{2r} = m_{2r}/m_2^r$. Then the optimal finite-dimensional upper constants in Question 5 for $p = 4$ and $p = 6$ are*

$$B_{4,q,n} = \frac{\beta_{4,q,n}}{\beta_{2,q,n}} \left(3 + \frac{\rho_4 - 3}{n} \right)^{1/4}$$

and

$$B_{6,q,n} = \frac{\beta_{6,q,n}}{\beta_{2,q,n}} \left(\frac{\rho_6 + 15(n-1)\rho_4 + 15(n-1)(n-2)}{n^2} \right)^{1/6}.$$

For $n \geq 2$, equality is attained exactly by the unit vectors satisfying $|a_1| = \dots = |a_n| = n^{-1/2}$. For $n = 1$ the statement is tautological.

Proof. The source paper's identity (2) gives, for every $p > -1$,

$$\left\| \sum_{i=1}^n a_i X_i \right\|_p = \beta_{p,q,n} \left\| \sum_{i=1}^n a_i Y_i \right\|_p,$$

where $Y_1, \dots, Y_n$ are independent copies of Y . Also

$$\left\| \sum_{i=1}^n a_i X_i \right\|_2 = \beta_{2,q,n} m_2^{1/2} \left(\sum_{i=1}^n a_i^2 \right)^{1/2}.$$

It is therefore enough to maximise $\|\sum a_i Y_i\|_p$ over $\sum a_i^2 = 1$.

Write $x_i = a_i^2$, so $x_i \geq 0$ and $\sum x_i = 1$. Odd mixed moments vanish. For $p = 4$,

$$\mathbb{E} \left(\sum_{i=1}^n a_i Y_i \right)^4 = m_4 \sum_i x_i^2 + 6m_2^2 \sum_{i < j} x_i x_j = 3m_2^2 + (m_4 - 3m_2^2) \sum_i x_i^2.$$

By Lemma 1, $m_4 - 3m_2^2 < 0$. Thus the fourth moment is maximised exactly when $\sum_i x_i^2$ is minimised on the simplex, namely at $x_i = 1/n$ for all i . The value at this point is

$$m_2^2 \left(3 + \frac{\rho_4 - 3}{n} \right).$$

After taking fourth roots and multiplying by $\beta_{4,q,n}/(\beta_{2,q,n} m_2^{1/2})$, the stated $B_{4,q,n}$ follows.

For $p = 6$, expansion gives

$$\mathbb{E} \left(\sum_{i=1}^n a_i Y_i \right)^6 = 15m_2^3 + c_2 \sum_i x_i^2 + c_3 \sum_i x_i^3,$$

where

$$c_2 = 15m_2(m_4 - 3m_2^2), \quad c_3 = m_6 - 15m_4m_2 + 30m_2^3.$$

We use the standard Schur criterion: a symmetric differentiable function F on the simplex is Schur-concave if

$$(x_i - x_j)(\partial_i F - \partial_j F) \leq 0$$

for all i, j . Here

$$(x_i - x_j)(\partial_i F - \partial_j F) = (x_i - x_j)^2 (2c_2 + 3c_3(x_i + x_j)).$$

By Lemma 1, $c_2 < 0$. If $c_3 \leq 0$, the bracket is strictly negative. If $c_3 > 0$, then $x_i + x_j \leq 1$, so

$$2c_2 + 3c_3(x_i + x_j) \leq 2c_2 + 3c_3 = 3m_6 - 15m_4m_2 < 0.$$

Thus the sixth moment is strictly Schur-concave in $x = (x_1, \dots, x_n)$. Since the uniform point $(1/n, \dots, 1/n)$ is majorised by every point of the simplex, it is the unique maximiser.

At $x_i = 1/n$, equivalently $a_i = \pm n^{-1/2}$, we have

$$\mathbb{E} \left(n^{-1/2} \sum_{i=1}^n Y_i \right)^6 = \frac{m_6 + 15(n-1)m_4m_2 + 15(n-1)(n-2)m_2^3}{n^2}.$$

Dividing by m_2^3 , taking sixth roots, and transferring back to X gives the displayed formula for $B_{6,q,n}$. $\square$

Verification Notes

The script `code/verify_p4_p6_moments.py` checks the gamma-function moment formulas

$$m_r = \frac{\Gamma((r+1)/q)}{\Gamma(1/q)}$$

for representative $q > 2$, verifies the inequalities $\rho_4 < 3$ and $\rho_6 < 5\rho_4$, checks the sixth-moment Schur bracket, and samples random simplex points to confirm that the equal point dominates for $p = 4$ and $p = 6$.

Limitations and Novelty Check

This is a partial answer to Question 5. It does not solve the missing finite-dimensional constants for arbitrary $p \geq 2$, nor the $p \in [1, 2]$ lower-constant side, nor the $p \in (-1, 1)$ problem mentioned immediately after the question.

Cheap run indexes were searched for arXiv:1801.07597 and the phrases “Sharp comparison of moments”, “ $B_{\{p,q,n\}}$ ”, “ $A_{\{p,q,n\}}$ ”, “Khintchine”, and “log-concave”. No existing packet was found for this source question. Web searches on 2026-07-18 used exact and nearby phrases including “ $B_{\{p,q,n\}} A_{\{p,q,n\}}$ Khintchine q_2^2 ”, “uniformly distributed on B_q optimal constants”, “ $p=4$ $B_{\{p,q,n\}} B_q$ ”, and “sixth moment uniformly distributed on B_q Khintchine”. These searches found the source paper and adjacent Khintchine/section literature, but no explicit resolution of the $p = 4$ or $p = 6$ finite- n subcases of Question 5.

References

References

- [1] A. Eskenazis, P. Nayar, and T. Tkocz, *Sharp comparison of moments and the log-concave moment problem*, Advances in Mathematics 334 (2018), 389–416; arXiv:1801.07597. <https://arxiv.org/abs/1801.07597>
- [2] A. W. Marshall and I. Olkin, *Inequalities: theory of majorization and its applications*, Academic Press, 1979.